DIF 0209 -

Integrales y técnicas de integración. Sustitución Trigonométrica

$\operatorname{Sen}^2(\theta) = 1 - \operatorname{Cos}^2(\theta)$

$\operatorname{Sec}^2(\theta) = \operatorname{Tan}^2(\theta) + 1$

$\operatorname{Tan}^2(\theta) = \operatorname{Sec}^2(\theta) - 1$

1.(a) $\displaystyle\int \frac{x^2}{\sqrt{1+x^2}}\,dx$

$x = \operatorname{Tan}(\theta)$,

$dx = \sec^2(\theta)\,d\theta$

$$= \int \frac{\operatorname{Tan}^2(\theta)}{\sqrt{1+\operatorname{Tan}^2(\theta)}}\sec^2(\theta)\,d\theta = \int \frac{\operatorname{Tan}^2(\theta)}{\sec(\theta)}\sec^2(\theta)\,d\theta = \underbrace{\int \operatorname{Tan}^2(\theta)\sec(\theta)\,d\theta}$$

$$= \int \frac{\operatorname{sen}^2(\theta)}{\cos^2(\theta)}\cdot\frac{1}{\cos(\theta)}\,d\theta = \int \operatorname{sen}^2(\theta)\cos^{-3}(\theta)\,d\theta$$

$$= \int (\sec^2(\theta)-1)\sec(\theta)\,d\theta$$

$$= \int \sec^3(\theta) - \sec(\theta)\,d\theta = \int \sec^3(\theta)\,d\theta - \int \sec(\theta)\,d\theta = \underbrace{\int \sec(\theta)\cdot\sec^2(\theta)\,d\theta} - \ln|\sec(\theta)+\operatorname{Ta}$$

$u = \sec(\theta)$ $\quad dv = \sec^2(\theta)\,d\theta$,

$du = \sec(\theta)\operatorname{Tan}(\theta)\,d\theta$ $\quad v = \operatorname{Tan}(\theta)$

$$\int \sec(\theta)\ldots$$

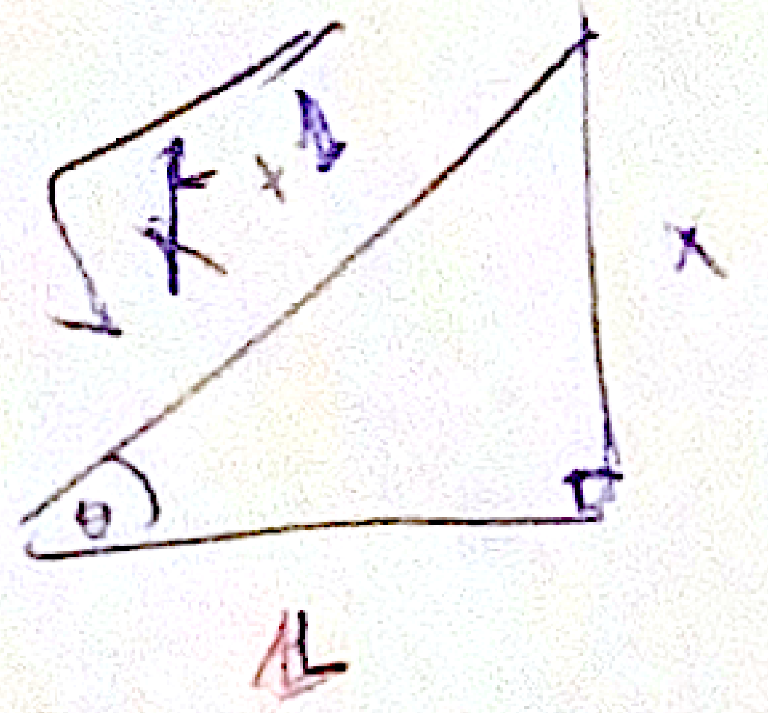
$$= \sec(\theta)\operatorname{Tan}(\theta) - \int \sec(\theta)\operatorname{Tan}^2(\theta)\,d\theta = \sec(\theta)\operatorname{Tan}(\theta) - \int \sec(\theta)(\sec^2(\theta)-1)\,d\theta$$

$$= \sec(\theta)\operatorname{Tan}(\theta) - \int \sec^3(\theta)\,d\theta - \int \sec(\theta)\,d\theta = \frac{\sec(\theta)\operatorname{Tan}(\theta)}{2} + \frac{1}{2}\left(\ln|\sec(\theta)+\operatorname{Tan}(\theta)|\right) - \ln|\sec(\theta)$$

$$= \frac{\sec(\theta)\operatorname{Tan}(\theta)}{2} - \frac{1}{2}\left(\ln|\sec(\theta)+\operatorname{Tan}(\theta)|\right) + C$$

$x = \operatorname{Tan}(\theta)$

$\xrightarrow{}$ 1



$$= \frac{\frac{\sqrt{x^2+1}}{1}\cdot x}{2} - \frac{1}{2}\ln\left|\frac{\sqrt{x^2+1}}{x} + x\right| + C$$

$$\int \sqrt{9-x^2}\,dx = \int \sqrt{9-9\,\text{sen}^2(\theta)} \cdot 3\cos(\theta)\,d\theta$$

$x = 3\,\text{sen}(\theta)$

$dx = 3\cos(\theta)\,d\theta$

$$= \int \sqrt{9(1-\text{sen}^2(\theta))} \cdot 3\cos(\theta)\,d\theta = 9\int \cos^2(\theta)\,d\theta$$

$$= 9\int \cos^2(\theta)\,d\theta$$

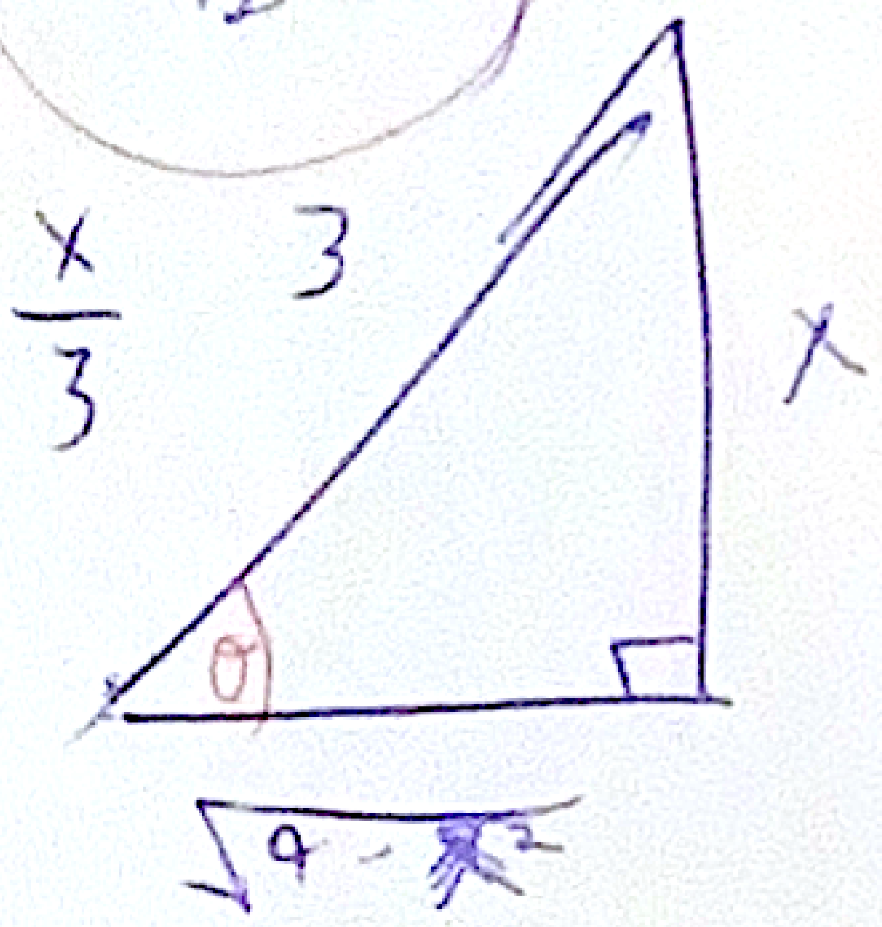
$$= 9\int \left(\frac{1}{2} + \frac{\cos(2\theta)}{2}\right) d\theta = \frac{9}{2}\int 1+\cos(2\theta)\,d\theta = \frac{9}{2}\left(\theta + \frac{\text{sen}(2\theta)}{2}\right) + C$$

$= \cos(\theta)\,\text{sen}(\theta)$

$$= \frac{9}{2}\left(\theta + \cos(\theta)\,\text{sen}(\theta)\right)$$

$$= \frac{9}{2}\left(\text{arcsen}\left(\frac{x}{3}\right) + \frac{x}{3} \cdot \frac{\sqrt{9-x^2}}{3}\right) + C$$

$\text{sen}(\theta) = \frac{x}{3}$



Integrales mediante descomposición en suma de fracciones parciales.

Solo para integrar divisiones de polinomios.

$$\int \frac{x}{x^2-5x+6}\,dx = \int \frac{x}{(x-2)(x-3)}\,dx = \int \left(\frac{3}{(x-3)} - \frac{2}{(x-2)}\right) dx$$

$$= 3\ln|x-3| - 2\ln|x-2|$$

$$A(x-2) + B(x-3) = x$$

Si $x=2$: $2 = -B$

Si $x=3$: $3 = A$

b) $\int \frac{x}{x^2+4x-21}\,dx = \int \frac{x}{(x+7)(x-3)}$

$x = \frac{A\cancel{(x+7)}(x-3)}{\cancel{x+7}} + \frac{B(x+7)\cancel{(x-3)}}{\cancel{x-3}}$

$x = A(x-3) + B(x+7)$

Si $x = 3$: $B(10)$

c) $\int \frac{5x-1}{(x+1)(x-3)}\,dx = \int \left(\frac{\frac{3}{2}}{x+1} + \frac{\frac{7}{2}}{x-3}\right)dx = \frac{3}{2}\left(\ln|(x+1)|\right) + \frac{7}{2}\ln|(x-3)| + C$

$5x-1 = A(x-3)+B(x+1)$

$x=3$: $14 = B(4) \Rightarrow \frac{7}{2} = B$

$x=-1$: $-6 = -A(4) \Rightarrow \frac{3}{2} = A$

$\int \frac{x}{(x-1)^2(x+1)}\,dx = \int \left(\frac{\frac{1}{4}}{(x-1)} + \frac{-\frac{1}{4}}{(x+1)} + \frac{\frac{1}{2}}{(x-1)^2}\right)dx = \frac{1}{4}\left(\ln|x-1|\right) - \frac{1}{4}\left(\ln|x+1|\right) + \frac{-1}{2(}$

$\frac{x}{(x-1)^2(x+1)} = \frac{A}{(x-1)} + \frac{B}{(x+1)} + \frac{C}{(x-1)^2}$

$x = A(x-1)(x+1) + B(x-1)^2 + C(x+1)$

Si $x = 1$: $1 = C(2) \Rightarrow \frac{1}{2} = C$

$-1 = B(-2)^2 \Rightarrow \frac{-1}{4} = B$

Si $x=0$: $0 = -A + B + C \Rightarrow A = B + C \Rightarrow \frac{1}{4} = A$

fa Caso de factor cuadrático irreducible.

ej: $\int \frac{1}{x(x^2+1)}\,dx$

$\ln|x| - \frac{1}{2}\int \frac{du}{u^2}$

$= \ln|x| - \frac{1}{2}\int u^{-2}\,du$

$=$

$\frac{A}{x} + \frac{\boxed{Bx+C}}{x^2+1} = \frac{1}{x(x^2+1)} \Big/ \cdot x(x^2+1)$

$\therefore \int = \int\left(\frac{1}{x} + \frac{-x}{(x^2+1)}\right)dx$

$1 = A(x^2+1) + (Bx+C)x$

$= \ln|x| - \int \frac{x}{x^2+1} = \ln|x| - \int \frac{du}{2u^2} = \ln|x| - 2\int u^{-1}\,du$

~~= [illegible]~~ $\checkmark - \ln|x|$

Si $X = 0$: $A = 1$

Si $x = 1$: $1 = 2 + B + C$ $\Rightarrow B + C = -1 \Rightarrow \underbrace{B = -C - 1}$

$-1 = B + C \longrightarrow$

$x = -1$: $1 = 2(\underset{1}{A}) + B - C = -1 \Rightarrow B - C = -1$

$-C - 1 - C = -1$

$-2C = 0$

$C = 0$

$B = -1 - 0$

$B = -1$